

LLM Engineering Study Map

Ed Donners' Udemy Course → AI Engineering from Scratch

A detailed topic-by-topic mapping of every concept in
Ed Donners' "AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents"
to the corresponding lessons in the free, open-source
AI Engineering from Scratch curriculum

Ed Donners Course	AI Engineering from Scratch
210 lectures, 33h 27m video	503 lessons, ~320h text+code
8 weeks, 8 projects	20 phases, 4 languages
Top-down: ship apps with APIs	Bottom-up: build from raw math
Paid (Udemy coupon ~\$12-15)	Free, open-source (MIT)
Python only	Python, TypeScript, Rust, Julia
330k students, 4.7/5 rating	47k GitHub stars, 150k readers

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Contents

1. How to Read This Document
2. Legend: Mapping Tags
3. Course Comparison at a Glance
4. Week 1 — Build Your First LLM Product: Exploring Top Models
5. Week 2 — Build a Multi-Modal Chatbot: LLMs, Gradio UI, and Agents
6. Week 3 — Open-Source Gen AI: Automated Solutions with HuggingFace
7. Week 4 — LLM Showdown: Evaluating Models for Code Gen & Business Tasks
8. Week 5 — Mastering RAG: Build Advanced Solutions with Vector Embeddings
9. Week 6 — From Traditional ML to DL to Fine-Tuning a Frontier Model
10. Week 7 — QLoRA Fine-Tuning Open-Source Models
11. Week 8 — Build Autonomous Multi-Agent System
12. Minimal From-Scratch Path (Priority Summary)
13. Recommended Study Strategy

1. How to Read This Document

This document maps every topic covered in Ed Donners' Udemy course "AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents" to the corresponding lessons in the free, open-source "AI Engineering from Scratch" curriculum by Rohit Ghumare (github.com/rohitg00/ai-engineering-from-scratch).

The two resources are complementary, not competing. Ed Donners teaches top-down: you ship 8 real LLM applications in 8 weeks using APIs and frameworks. AI Engineering from Scratch teaches bottom-up: you build backpropagation, tokenizers, attention mechanisms, and agent loops from raw math before any framework gets imported.

Since you already know the fundamentals (Python, basic ML, basic DL), this mapping skips Phases 0-2 (Setup, Math Foundations, ML Fundamentals) and Phase 3 basics. It focuses on Phases 5, 7, 10, 11, 13, 14, 15, 16, 17, and 19 — the layers that give you the "why" behind what Ed Donners teaches you to build.

2. Legend: Mapping Tags

[CORE] — The from-scratch lesson that directly teaches the fundamental concept behind what Ed Donners covers. Study this alongside the corresponding Ed Donners week.

[DEEP] — Goes beyond what Ed covers. Study this if you want interview-level depth or are building production systems. Optional but recommended.

[SKIP] — You already know this (environment setup, basic Python, etc.). Listed so you know it exists; do not study unless you need a refresher.

Phase Reference (AI Engineering from Scratch):

Phase	Name	Lessons
0	Setup & Tooling	12
1	Math Foundations	22
2	ML Fundamentals	18
3	Deep Learning Core	13
4	Computer Vision	28
5	NLP: Foundations to Advanced	29
6	Speech & Audio	17
7	Transformers Deep Dive	16
8	Generative AI	15
9	Reinforcement Learning	12
10	LLMs from Scratch	24
11	LLM Engineering	17
12	Multimodal AI	25
13	Tools & Protocols	23
14	Agent Engineering	42
15	Autonomous Systems	22
16	Multi-Agent & Swarms	—
17	Infrastructure & Production	28

18	Ethics, Safety & Alignment	30
19	Capstone Projects	85

4. Week 1 — Build Your First LLM Product: Exploring Top Models

37 lectures, 5h 46min. Covers Ollama, OpenAI API, frontier model comparison, transformers overview, tokenization, and the brochure generator project.

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
Ollama, local LLMs, open-source models	P10-L14 Open Models: Architecture Walkthroughs [CORE] P10-L11 Quantization: INT8, GPTQ, AWQ, GGUF [DEEP] — understand why Ollama can run models locally; quantization is the enabler	CORE
OpenAI API, chat completions, system vs user prompts	P11-L01 Prompt Engineering: Techniques and Patterns [CORE] P11-L05 Context Engineering [DEEP] — context window management beyond simple API calls	CORE
Frontier models: GPT, Claude, Gemini, Grok compared	P10-L14 Open Models: Architecture Walkthroughs [CORE] P10-L20 DeepSeek-V3 Architecture Walkthrough [DEEP] P7-L13 Scaling Laws [DEEP] — why frontier models are big; the science behind model selection	CORE
Transformers architecture overview	P7-L01 Why Transformers: Problems with RNNs [CORE] P7-L02 Self-Attention from Scratch [CORE] P7-L05 The Full Transformer: Encoder + Decoder [CORE] P7-L14 Build a Transformer from Scratch [DEEP] — Ed explains conceptually; here you build one	CORE
Tokens, tokenization, tiktoken, context windows, API costs	P10-L01 Tokenizers: BPE, WordPiece, SentencePiece [CORE] P10-L02 Building a Tokenizer from Scratch [DEEP] — build BPE yourself; Ed only uses tiktoken P5-L19 Subword Tokenization [CORE] P11-L11 Caching, Rate Limiting and Cost [DEEP]	CORE
Base, chat, reasoning model types	P7-L07 GPT: Causal Language Modeling [CORE] P10-L06 Instruction Tuning: SFT [CORE] — what makes a chat model vs base model P10-L07 RLHF: Reward Model + PPO [DEEP] P10-L08 DPO [DEEP]	CORE
Chaining LLM calls, brochure generator (Project 1)	P11-L01 Prompt Engineering [CORE] P11-L05 Context Engineering [CORE] — chaining is context engineering P14-L01 The Agent Loop [DEEP] — chaining calls is a proto-agent-loop	CORE
Dev environment setup (Cursor, UV, Git)	P0-L01 Dev Environment [SKIP] P0-L04 APIs and Keys [SKIP] P0-L08 Editor Setup [SKIP]	SKIP

5. Week 2 — Build a Multi-Modal Chatbot: LLMs, Gradio UI, and Agents

24 lectures, 3h 41min. Covers multi-model conversations, Gradio UIs, tool calling, agentic AI intro, DALL-E, TTS, and the airline assistant project.

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
Multi-model conversations (OpenAI + Claude)	P11-L17 Agent Framework Tradeoffs [CORE] P13-L21 LLM Routing Layer [DEEP] — routing between models is a production pattern	CORE
Gradio UIs, streaming, chat interfaces	P11-L13 Building a Production LLM App [CORE] — covers the app layer but not Gradio specifically. Ed Donners is stronger here.	CORE
System prompts, multi-shot prompting, first look at RAG	P11-L01 Prompt Engineering [CORE] P11-L02 Few-Shot, CoT, Tree-of-Thought [CORE] P11-L06 RAG: Retrieval-Augmented Generation [CORE]	CORE
Tool calling / function calling	P11-L09 Function Calling and Tool Use [CORE] P13-L01 The Tool Interface [CORE] P13-L02 Function Calling Deep Dive [DEEP] — the mechanics behind the prompt-injected tool pattern Ed teaches P13-L03 Parallel and Streaming Tool Calls [DEEP]	CORE
Airline AI assistant with tool calling + SQLite (Project 2)	P13-L05 Tool Schema Design [CORE] P14-L06 Tool Use and Function Calling [DEEP] — agent-level tool use, not just single-call	CORE
Agentic AI intro, multi-tool workflows	P14-L01 The Agent Loop [CORE] P14-L02 ReWOO and Plan-and-Execute [CORE] P14-L12 Anthropic's Workflow Patterns [CORE] — the canonical patterns Ed is introducing you to	CORE
DALL-E 3, text-to-speech, multimodal	P8-L07 Latent Diffusion and Stable Diffusion [CORE] P8-L06 Diffusion Models: DDPM from Scratch [DEEP] — DALL-E is a diffusion model; understand it from scratch P6-L07 Text-to-Speech (TTS) [CORE] P12-L01 Vision Transformers and the Patch-Token Primitive [DEEP]	CORE
Multimodal AI assistant (Project 2 continued)	P12-L05 LLaVA and Visual Instruction Tuning [CORE] P12-L25 Multimodal Agents and Computer-Use [DEEP]	CORE

6. Week 3 — Open-Source Gen AI: Automated Solutions with HuggingFace

23 lectures, 3h 36min. Covers HuggingFace platform, Google Colab, HF Pipelines, tokenizers in action, transformers low-level API, and inside LLaMA architecture.

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
HuggingFace platform: models, datasets, Spaces	P10-L14 Open Models: Architecture Walkthroughs [CORE] — covers the model landscape you navigate on HF	CORE
Google Colab, cloud GPUs	P0-L03 GPU Setup and Cloud [SKIP]	SKIP
HF Pipelines (sentiment, NER, Q&A;, image, audio, diffusion)	P5-L05 Sentiment Analysis [CORE] P5-L06 Named Entity Recognition (NER) [CORE] P5-L13 Question Answering Systems [CORE] P8-L07 Latent Diffusion and Stable Diffusion [CORE] — what pipeline('text2image') does underneath	CORE
Tokenizers in action: encoding/decoding, chat templates, special tokens	P10-L01 Tokenizers: BPE, WordPiece, SentencePiece [CORE] P10-L02 Building a Tokenizer from Scratch [DEEP] P5-L19 Subword Tokenization [CORE] P5-L20 Structured Outputs and Constrained Decoding [DEEP] — chat templates = constrained decoding	CORE
Transformers low-level API, quantization	P10-L11 Quantization: INT8, GPTQ, AWQ, GGUF [CORE] P10-L12 Inference Optimization [DEEP] P7-L12 KV Cache, Flash Attention and Inference Optimization [DEEP] — what model.generate() does under the hood	CORE
Inside LLaMA: PyTorch model architecture, token embeddings	P7-L05 The Full Transformer [CORE] P7-L07 GPT: Causal Language Modeling [CORE] P10-L04 Pre-Training a Mini GPT (124M) [DEEP] — build the architecture you are inspecting in LLaMA P10-L14 Open Models: Architecture Walkthroughs [CORE]	CORE
Decoder layers, attention, non-linearity	P7-L02 Self-Attention from Scratch [CORE] P7-L03 Multi-Head Attention [CORE] P3-L04 Activation Functions: ReLU, Sigmoid, GELU and Why [CORE] — the why non-linearity matters Ed mentions	CORE
Running Phi, Gemma, Qwen, DeepSeek with HF	P10-L14 Open Models: Architecture Walkthroughs [CORE] P10-L20 DeepSeek-V3 Architecture Walkthrough [DEEP]	CORE
Token-by-token inference visualization	P7-L12 KV Cache, Flash Attention and Inference Optimization [CORE] P10-L12 Inference Optimization [DEEP] P7-L16 Speculative Decoding [DEEP]	CORE

7. Week 4 — LLM Showdown: Evaluating Models for Code Gen and Business Tasks

21 lectures, 3h 9min. Covers model evaluation, code generation, reasoning models, business task evaluation, and the LLM competition game.

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
Model evaluation, comparing frontier models on tasks	P10-L10 Evaluation: Benchmarks, Evals [CORE] P5-L27 LLM Evaluation: RAGAS, DeepEval, G-Eval [CORE] P5-L28 Long-Context Evaluation: NIAH, RULER, LongBench [DEEP]	CORE
Code generation (Python to C++, Project 4)	P14-L01 The Agent Loop [CORE] — code gen as an agent task P19-L01 Terminal-Native Coding Agent (capstone) [DEEP] P19-L09 Code Migration Agent [DEEP] — repo-level code transformation	CORE
Reasoning models, reasoning effort	P11-L02 Few-Shot, CoT, Tree-of-Thought [CORE] — reasoning = CoT at scale P14-L04 Tree of Thoughts and LATS [DEEP] P15-L02 STaR, V-STaR, Quiet-STaR: Self-Taught Reasoning [DEEP]	CORE
Business task evaluation, model selection	P11-L10 Evaluation and Testing [CORE] P11-L17 Agent Framework Tradeoffs [CORE] P13-L21 LLM Routing Layer [DEEP] — model selection as routing	CORE
Agentic AI in action: Deep Research, Claude Code, Agent Mode	P14-L01 The Agent Loop [CORE] P14-L12 Anthropic's Workflow Patterns [CORE] P15-L01 From Chatbots to Long-Horizon Agents [DEEP]	CORE
LLM competition game	P10-L10 Evaluation: Benchmarks, Evals [CORE] — what you are really doing is building an eval harness	CORE

8. Week 5 — Mastering RAG: Build Advanced Solutions with Vector Embeddings

32 lectures, 5h 25min. The most RAG-heavy week. Covers vector data stores, LangChain, Chroma, FAISS, chunking, RAG pipelines, evaluation (MRR, NDCG, LLM-as-judge), advanced RAG (query rewriting, re-ranking, GraphRAG), and the knowledge worker project.

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
RAG big idea, vector data stores	P11-L06 RAG: Retrieval-Augmented Generation [CORE] — the canonical from-scratch RAG lesson	CORE
Vectors, LangChain, vector databases	P11-L04 Embeddings and Vector Representations [CORE] P5-L03 Word Embeddings: Word2Vec from Scratch [DEEP] — build embeddings from raw math; Ed only uses pre-trained ones P5-L22 Embedding Models Deep Dive [CORE]	CORE
Document chunking, text splitters	P5-L23 Chunking Strategies for RAG [CORE] P11-L07 Advanced RAG: Chunking, Reranking [CORE] P19-L64 Chunking Strategies, Compared (capstone) [DEEP]	CORE
Encoder models vs vector databases, Chroma, FAISS	P7-L06 BERT: Masked Language Modeling [CORE] — encoder models = BERT family P5-L14 Information Retrieval and Search [CORE] — the IR theory behind vector DBs	CORE
Embedding visualization with t-SNE	P1-L10 Dimensionality Reduction: PCA, t-SNE, UMAP [CORE] — Ed uses t-SNE as a black box; understand it from scratch	CORE
Complete RAG pipeline with LangChain + Chroma	P11-L06 RAG: Retrieval-Augmented Generation [CORE] P11-L07 Advanced RAG: Chunking, Reranking [CORE] P19-L69 End-to-End RAG System (capstone) [DEEP]	CORE
RAG with conversation history, Gradio UI	P5-L29 Dialogue State Tracking [CORE] P11-L05 Context Engineering [CORE] — conversation history = context management	CORE
RAG evaluation: MRR, NDCG, LLM-as-judge, golden data	P5-L27 LLM Evaluation: RAGAS, DeepEval, G-Eval [CORE] P19-L68 RAG Evaluation: Precision, Recall, MRR, nDCG, Faithfulness [DEEP] — build the eval harness Ed uses	CORE
Advanced RAG: query rewriting, re-ranking, GraphRAG	P11-L07 Advanced RAG: Chunking, Reranking [CORE] P19-L67 Query Rewriting: HyDE, Multi-Query, and Decomposition [DEEP] P19-L66 Cross-Encoder Reranker [DEEP] P19-L65 Hybrid Retrieval with BM25 and Dense Embeddings [DEEP] P5-L26 Relation Extraction and Knowledge Graph Construction [DEEP] — GraphRAG foundation	CORE
RAG without LangChain, semantic chunking	P19-L69 End-to-End RAG System [DEEP] — builds RAG without frameworks P5-L23 Chunking Strategies for RAG [CORE]	CORE
Knowledge worker (Project 5)	P19-L08 Production RAG Chatbot (Regulated Vertical) [DEEP] P19-L02 RAG over Codebase [DEEP]	CORE

9. Week 6 — From Traditional ML to DL to Fine-Tuning a Frontier Model

27 lectures, 4h 47min. Covers dataset curation, traditional ML baselines, neural networks in PyTorch, testing frontier models, and fine-tuning GPT-4o-mini with OpenAI's SFT API.

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
Datasets, curation, Amazon data, HF datasets	P10-L03 Data Pipelines for Pre-Training [CORE] P2-L08 Feature Engineering and Selection [CORE]	CORE
Dataset distribution, deduplication, weighted sampling	P1-L15 Statistics for ML [CORE] P1-L16 Sampling Methods [CORE] P2-L09 Model Evaluation: Metrics, Cross-Validation [CORE]	CORE
Five-step AI process, MLOps, productionizing	P11-L13 Building a Production LLM App [CORE] P17 (entire phase) Infrastructure and Production [DEEP] — the full production stack: serving, observability, FinOps, compliance	CORE
Batch processing with Groq API, JSONL	P11-L11 Caching, Rate Limiting and Cost [CORE] P17-L15 Batch APIs: the 50% Discount [DEEP]	CORE
Traditional ML baselines: Random Forest, XGBoost	P2-L04 Decision Trees and Random Forests [CORE] P2-L11 Ensemble Methods: Boosting, Bagging, Stacking [CORE] — XGBoost = gradient boosting	CORE
Bag of Words, CountVectorizer, linear regression for NLP	P5-L02 Bag of Words, TF-IDF and Text Representation [CORE] P2-L02 Linear Regression from Scratch [CORE]	CORE
Neural networks with PyTorch	P3-L11 Introduction to PyTorch [CORE] P3-L02 Multi-Layer Networks and Forward Pass [CORE] P3-L03 Backpropagation from Scratch [DEEP] — Ed uses PyTorch; here you build backprop by hand first	CORE
Testing frontier models on price prediction (Project 6)	P11-L10 Evaluation and Testing [CORE] P10-L10 Evaluation: Benchmarks, Evals [CORE]	CORE
Fine-tuning GPT-4o-mini with OpenAI SFT API	P10-L06 Instruction Tuning: SFT [CORE] — the theory behind what OpenAI's API does P11-L08 Fine-Tuning with LoRA and QLoRA [CORE]	CORE
When fine-tuning fails, deep neural network redemption	P3-L07 Regularization: Dropout, Weight Decay, BatchNorm [CORE] — why models overfit P3-L13 Debugging Neural Networks [CORE] P2-L10 Bias, Variance and the Learning Curve [CORE]	CORE

10. Week 7 — QLoRA Fine-Tuning Open-Source Models

Covers LoRA, QLoRA quantization, hyperparameters, dataset prep, base vs chat models, Weights & Biases, training monitoring, cross-entropy loss, and inference on fine-tuned models.

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
LoRA: training LLaMA with low-rank adapters	P11-L08 Fine-Tuning with LoRA and QLoRA [CORE] P8-L08 ControlNet, LoRA and Conditioning [DEEP] — LoRA originated in diffusion models; understand the full lineage	CORE
QLoRA quantization explained, 8-bit/4-bit loading	P10-L11 Quantization: INT8, GPTQ, AWQ, GGUF [CORE] — the quantization theory QLoRA relies on P17-L09 Production Quantization: GPTQ, AWQ, GGUF, FP8, NVFP4 [DEEP]	CORE
LoRA hyperparameters, parameter calculations	P2-L12 Hyperparameter Tuning [CORE] P3-L09 Learning Rate Schedules and Warmup [CORE]	CORE
Dataset prep for fine-tuning, token length optimization	P10-L03 Data Pipelines for Pre-Training [CORE] P10-L06 Instruction Tuning: SFT [CORE] — SFT data formatting P5-L19 Subword Tokenization [CORE] — token length = vocab design	CORE
Base models vs chat models, LLaMA fine-tuning	P7-L07 GPT: Causal Language Modeling [CORE] — base model P10-L06 Instruction Tuning: SFT [CORE] — chat model = base + SFT + RLHF	CORE
Training hyperparameters: learning rate, optimizers	P3-L06 Optimizers: SGD, Momentum, Adam, AdamW [CORE] P3-L09 Learning Rate Schedules and Warmup [CORE]	CORE
Weights & Biases, HF SFT Trainer, TRL	P2-L13 ML Pipelines and Experiment Tracking [CORE] — W&B; = experiment tracking P10-L05 Distributed Training, FSDP, DeepSpeed [DEEP]	CORE
Monitoring training loss, overfitting, checkpoint selection	P3-L05 Loss Functions: MSE, Cross-Entropy, Contrastive [CORE] P3-L13 Debugging Neural Networks [CORE] P2-L10 Bias, Variance and the Learning Curve [CORE]	CORE
Cross-entropy loss, probability distributions	P3-L05 Loss Functions: MSE, Cross-Entropy, Contrastive [CORE] P1-L09 Information Theory: Entropy, KL Divergence [DEEP] — the math behind cross-entropy	CORE
Inference on fine-tuned model, comparing with frontier (Project 7)	P10-L12 Inference Optimization [CORE] P10-L10 Evaluation: Benchmarks, Evals [CORE] P19-L07 End-to-End Fine-Tuning Pipeline [DEEP]	CORE

11. Week 8 — Build Autonomous Multi-Agent System

22 lectures, 3h 6min. Covers agentic AI, agent architectures, serverless deployment on Modal, multi-agent collaboration, and the autonomous deal-spotting system (Project 8).

Ed Donners Topic	AI Engineering from Scratch Mapping	Tag
Agentic AI intro, agent architectures	P14-L01 The Agent Loop [CORE] P14-L12 Anthropic's Workflow Patterns [CORE] P14-L28 Orchestration Patterns: Supervisor, Swarm, Hierarchical [CORE]	CORE
Serverless deployment on Modal	P17 (entire phase) Infrastructure and Production [DEEP] — Modal is one option; understand the full serving landscape P17-L10 Cold Start Mitigation for Serverless LLMs [DEEP]	CORE
Designing agent architectures	P14-L02 ReWOO and Plan-and-Execute [CORE] P14-L03 Reflexion and Verbal RL [CORE] P14-L04 Tree of Thoughts and LATS [CORE] P14-L11 Planning with HTN and Evolutionary Search [DEEP]	CORE
Multi-agent collaboration, deal-spotting (Project 8)	P14-L25 Multi-Agent Debate and Collaboration [CORE] P14-L15 Role-Based Agent Teams: CrewAI [CORE] P16 Multi-Agent and Swarms (entire phase) [DEEP] P19-L10 Multi-Agent Software Engineering Team [DEEP]	CORE
Autonomous systems, self-improvement	P15 Autonomous Systems (entire phase, 22 lessons) [DEEP] P15-L01 From Chatbots to Long-Horizon Agents [CORE] P15-L02 STaR, V-STaR, Quiet-STaR [DEEP]	CORE
Agent memory	P14-L07 Agent Memory: Virtual Context and Memory Paging [CORE] P14-L08 Memory Blocks and Sleep-Time Compute [DEEP] P14-L09 Hybrid Memory: Vector + Graph + KV [DEEP]	CORE
Agent failure modes, safety	P14-L26 Failure Modes: Why Agents Break [CORE] P14-L27 Prompt Injection and the PVE Defense [CORE] P18-L15 Indirect Prompt Injection [DEEP]	CORE
Agent observability, production	P14-L23 OpenTelemetry GenAI Semantic Conventions [CORE] P14-L24 Agent Observability: Langfuse, Phoenix, Opik [CORE] P14-L29 Production Runtimes [DEEP]	CORE

12. Minimal From-Scratch Path (Priority Summary)

Since you already know the fundamentals, here is the shortest from-scratch path that fills every gap Ed Donners leaves. Listed in the order you would encounter them while doing Ed's course. Each entry can be studied before or after the corresponding Ed Donners week.

#	From-Scratch Lessons	Why	Est. Time
1	P7-L02, L03, L05, L14 (Transformers)	Build attention + full transformer by hand. Ed only explains conceptually.	~8h
2	P10-L01, L02 (Tokenizers)	Build BPE from scratch. Ed only uses tiktoken.	~4h
3	P10-L06, L07, L08 (SFT, RLHF, DPO)	Understand what makes chat models chat. Ed uses APIs.	~6h
4	P11-L06, L07 (RAG + Advanced RAG)	Build RAG from first principles. Ed uses LangChain.	~4h
5	P5-L23, P19-L64 to L69 (Chunking + RAG Capstone)	Build RAG engineering Ed hints at but does not build.	~8h
6	P10-L11 (Quantization)	Understand why QLoRA works. Ed uses it as a black box.	~3h
7	P3-L03 (Backprop from Scratch)	If you have not written backprop by hand, do it once.	~3h
8	P14-L01, L02, L12 (Agent Loop, Plan-Execute, Workflow Patterns)	Build Week 8 multi-agent system.	~4h
9	P14-L25, L28 (Multi-Agent + Orchestration)	Patterns for Ed's capstone Project 8.	~3h
10	P10-L10, P5-L27 (Evaluation)	Build eval harnesses instead of eyeballing results.	~4h

Total: ~47 hours of from-scratch lessons to get the "why" behind all 33 hours of Ed Donners' "what". Do them in parallel: each week of Ed, spend ~6h on the corresponding from-scratch lessons before or after.

13. Recommended Study Strategy

Phase 1: Ship First (Weeks 1-4 with Ed Donners)

Go through Ed Donners' Weeks 1-4 as-is. Build the brochure generator, the airline assistant, the meeting minutes tool, and the code optimizer. These are your portfolio pieces. In parallel, study from-scratch lessons #1 and #2 (Transformers + Tokenizers) so you understand what the models you are calling actually do internally.

Phase 2: Go Deep on RAG (Week 5 with both)

Ed's Week 5 is the most RAG-intensive week. Before starting it, study from-scratch lessons #4 and #5 (RAG from first principles + chunking/RAG capstone track). Then do Ed's Week 5 with full understanding of what LangChain and Chroma are doing under the hood. This is where the from-scratch curriculum adds the most value.

Phase 3: Fine-Tuning Theory + Practice (Weeks 6-7 with both)

Before Ed's Week 6, study from-scratch lesson #7 (Backprop from Scratch) if you have not written it by hand. Before Ed's Week 7 (QLoRA), study from-scratch lesson #6 (Quantization) and #3 (SFT, RLHF, DPO). Then Ed's QLoRA week becomes much clearer: you understand why 4-bit quantization works, what LoRA adapters are mathematically, and what the SFT trainer is doing to the model.

Phase 4: Agents (Week 8 with both)

Before Ed's Week 8, study from-scratch lessons #8 and #9 (Agent Loop, Plan-Execute, Workflow Patterns, Multi-Agent + Orchestration). Then Ed's multi-agent deal-spotter project becomes a concrete implementation of patterns you already understand theoretically.

Phase 5: Evaluation (Throughout)

Study from-scratch lesson #10 (Evaluation) early. Ed evaluates models by eyeballing results and running ad-hoc comparisons. The from-scratch curriculum teaches you to build proper eval harnesses with MRR, nDCG, RAGAS, and LLM-as-judge. This is the single most important skill for interviews and production work that Ed's course undercovers.

Resources

Ed Donners course: <https://www.udemy.com/course/llm-engineering-master-ai-and-large-language-models/>

AI Engineering from Scratch: <https://aiengineeringfromscratch.com>

GitHub repo (clone or read lessons): <https://github.com/rohitg00/ai-engineering-from-scratch>

Interactive terminal learning: Run `npx skills add rohitg00/ai-engineering-from-scratch` then `/start - learning` in Claude Code, Cursor, or Codex for a placement quiz and personalized path.